MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

PEDIATRIC HEAD AND NECK SURGERY ANTIMICROBIAL PROPHYLAXIS RECOMMENDATIONS

Nature of Procedure	Recommended Regimen	Alternative Regimen	Post-operative duration[1]
Clean procedures *(incision does not transect oral, pharyngeal, or nasal mucosa)*			
No prosthesis	**Antibiotics Not Recommended**		N/A
Placement of prosthesis *(e.g. BAHA device placement, TMJ prosthesis, NOT tympanostomy tubes)*	**Cefazolin**	High-risk cephalosporin or severe non-IgE beta-lactam allergy: **Clindamycin**	None *Perforation of oral mucosa or auditory canal: 5 days amp-sulb / amox-clav*
Mandibular Distractions			≤ 48 Hours
Tracheobronchial Splints *without tracheal violation*			48 Hours
Skull Base (Lateral or Posterior) (including cochlear implants)	**Cefuroxime**	High-risk cephalosporin or severe non-IgE beta-lactam allergy: **Clindamycin**	None
Clean-contaminated procedures *(incision transects oral, pharyngeal, or nasal mucosa)*			
Clean-Contaminated *(NOT tonsillectomy, tracheostomy, FESS, or middle ear/mastoid case [unless active drainage])*	**Ampicillin/sulbactam**	Penicillin Allergy: **Cefazolin + Metronidazole**	None
Operative Facial Fractures[2] *(Mandibular & non-mandibular)*			≤ 24 Hours *Perioperative only, w/ each procedure*
Tracheobronchial Splints *with tracheal violation*			7 Days
Facial osteotomy *w/ primary bone grafting*			72 Hours
Non-operative facial fractures, basilar skull fractures, CSF leak, or nasal packing	**Antibiotics Not Recommended**		N/A
Skull Base (Anterior)	**Ampicillin/sulbactam**	Penicillin Allergy: **Cefuroxime + Metronidazole**	*Endoscopic pituitary adenomas:* None **lumbar drain/EVD prophylaxis per neurosurgery* *All other procedures:* ≤72 Hours*
	Add **vancomycin** to above if history of MRSA infection or colonization in prior 12 months.		
Airway Reconstruction *includes resection (tracheal resection, cricotracheal resection, bronchial resection), slide (slide tracheoplasty), and augmentation (laryngotracheal reconstruction, cricoid split)*	**Ampicillin/sulbactam**; History of *Pseudomonas aeruginosa* in prior 12 months OR a current tracheostomy without cultures in prior 3 months: **Piperacillin/tazobactam**	Penicillin Allergy: **Cefuroxime + Metronidazole;** History of *Pseudomonas aeruginosa* in prior 12 months OR a current tracheostomy without cultures in prior 3 months: **Cefepime + Metronidazole**	72 Hours
	Add **vancomycin** to above history of MRSA infection or colonization in prior 12 months OR posterior cricoid split without cultures in prior 3 months.		

Footnotes: 1. May transition to enteral equivalent antibiotic (ie: cefazolin → cephalexin or ampicillin/sulbactam → amoxicillin/clavulanate) once safe for enteral route. 2. For patients with chronic open wounds awaiting definitive soft tissue wound closure, empiric prophylactic antibiotic therapy is not indicated. If there is concern for infection, ID consultation is recommended. Abbreviations- amp-sulb = ampicillin-sulbactam; amox-clav = amoxicillin-clavulanic acid

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References

	Citation
12/2025	Berríos-Torres SI, Umscheid CA, Bratzler DW, et al; Healthcare Infection Control Practices Advisory Committee. Centers for Disease Control and Prevention guideline for the prevention of surgical site infection, 2017. JAMA Surg. 2017;152(8):784–791. doi:10.1001/jamasurg.2017.0904

Revision History

Date	Revision
12/2025	Created New Document

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.